

DS 522 Data Acquisition and Analytics

HOS07A-2 Data Transformation with PBI

Please capture screenshots from the steps indicated below when performing HOS07A-2 and paste them in this document. **Save this document as a PDF for submission.**

STEP1: Creating Calculated Columns using **SALES** table

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with columns: InetKey, StoreKey, ProductKey, PromotionKey, UnitCost, UnitPrice, SalesQuantity, ReturnQuantity, ReturnAmount, DiscountQuantity, DiscountAmount, TotalCost, SalesAmount, Margin, and Margin Percent. The table contains 170 rows of data. The right-hand pane shows the 'Data' view with a search bar and a list of fields. The 'Sales' table is selected, and the fields list includes: ProductCategory, ProductSubcategory, UnitCost, UnitPrice, Weight, Weight Label, WeightUnitMeasureID, ProductCategory, ProductSubcategory, Sales, channelKey, DateKey, DiscountAmount, DiscountQuantity, Margin, Margin Percent, ProductKey, PromotionKey, ReturnAmount, ReturnQuantity, SalesAmount, SalesKey, SalesQuantity, StoreKey, TotalCost, UnitCost, and UnitPrice. The bottom status bar indicates 'Table: Sales (3,406,089 rows)' and the system clock shows 7:26 PM on 5/21/2026.

InetKey	StoreKey	ProductKey	PromotionKey	UnitCost	UnitPrice	SalesQuantity	ReturnQuantity	ReturnAmount	DiscountQuantity	DiscountAmount	TotalCost	SalesAmount	Margin	Margin Percent
1	117	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	38	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	196	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	275	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	124	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	232	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	269	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	129	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	38	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	15	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	62	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	93	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	2	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	171	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	249	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	69	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	272	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	176	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	237	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	299	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	18	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	43	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	198	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	153	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	272	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	14	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	150	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	222	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	21	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	77	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	122	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	83	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	57	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203
1	170	1930	1	\$152.94	\$299.99	10	0	\$0	0	\$0	\$1,529.4	\$2,999.9	\$1,470.5	0.490183006100203

STEP2: Adding Date functions in a DAX Expression using **CALENDAR** table

Table: Calendar (2,556 rows)

Date	Weekday
Saturday, January 1, 2005	Sat
Sunday, January 2, 2005	Sun
Monday, January 3, 2005	Mon
Tuesday, January 4, 2005	Tue
Wednesday, January 5, 2005	Wed
Thursday, January 6, 2005	Thu
Friday, January 7, 2005	Fri
Saturday, January 8, 2005	Sat
Sunday, January 9, 2005	Sun
Monday, January 10, 2005	Mon
Tuesday, January 11, 2005	Tue
Wednesday, January 12, 2005	Wed
Thursday, January 13, 2005	Thu
Friday, January 14, 2005	Fri
Saturday, January 15, 2005	Sat
Sunday, January 16, 2005	Sun
Monday, January 17, 2005	Mon
Tuesday, January 18, 2005	Tue
Wednesday, January 19, 2005	Wed
Thursday, January 20, 2005	Thu
Friday, January 21, 2005	Fri
Saturday, January 22, 2005	Sat
Sunday, January 23, 2005	Sun
Monday, January 24, 2005	Mon
Tuesday, January 25, 2005	Tue
Wednesday, January 26, 2005	Wed
Thursday, January 27, 2005	Thu
Friday, January 28, 2005	Fri
Saturday, January 29, 2005	Sat
Sunday, January 30, 2005	Sun
Monday, January 31, 2005	Mon
Tuesday, February 1, 2005	Tue
Wednesday, February 2, 2005	Wed
Thursday, February 3, 2005	Thu

STEP3: Using Data from Related table in an Expression using **PRODUCT** table

Table: Product (2,517 rows)

Option	Manufacturer	BrandName	ClassName	UnitCost	UnitPrice	ProductKey	ProductSubcategoryKey	Weight	WeightUnitMeasureID	Weight Label	Product Category
photos and text, display JPEG, BMP, GIF, TIFF and PNG	Cortoso, Ltd	Cortoso	Regular	\$61.62	\$134	20		1	pounds	1 pounds	Audio
ry life	Cortoso, Ltd	Cortoso	Regular	\$84.49	\$255	29		1	1.2 pounds	1.2 pounds	Audio
ry life	Cortoso, Ltd	Cortoso	Regular	\$84.49	\$255	31		1	1.2 pounds	1.2 pounds	Audio
1A	Cortoso, Ltd	Cortoso	Regular	\$48.92	\$95.95	36		1	2.9 pounds	2.9 pounds	Audio
photos and text, share JPEG, BMP, GIF, TIFF	Cortoso, Ltd	Cortoso	Regular	\$106.69	\$232	44		1	1 pounds	1 pounds	Audio
1 subwoofer and 1 A/V receiver	Cortoso, Ltd	Cortoso	Regular	\$167.73	\$329	241		11	44.1 pounds	44.1 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$294.54	\$889	242		11	11.4 pounds	11.4 pounds	TV and Video
VD or USB	Cortoso, Ltd	Cortoso	Regular	\$208.52	\$409	243		11	37.2 pounds	37.2 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$197.28	\$429	244		11	19 pounds	19 pounds	TV and Video
with 5.1-channel DVD players, computer systems, and more	Cortoso, Ltd	Cortoso	Regular	\$224.87	\$489	245		11	41 pounds	41 pounds	TV and Video
il in/out	Cortoso, Ltd	Cortoso	Regular	\$167.73	\$329	246		11	7.5 pounds	7.5 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$155.89	\$339	247		11	11.4 pounds	11.4 pounds	TV and Video
ubwoofer	Cortoso, Ltd	Cortoso	Regular	\$183.49	\$399	248		11	65 pounds	65 pounds	TV and Video
ubwoofer	Cortoso, Ltd	Cortoso	Regular	\$243.27	\$529	250		11	30.5 pounds	30.5 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$244.72	\$480	251		11	32 pounds	32 pounds	TV and Video
id 1 subwoofer	Cortoso, Ltd	Cortoso	Regular	\$183.54	\$360	252		11	73 pounds	73 pounds	TV and Video
subwoofer and 1 A/V receiver	Cortoso, Ltd	Cortoso	Regular	\$167.73	\$329	254		11	32.3 pounds	32.3 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$294.54	\$889	255		11	37.5 pounds	37.5 pounds	TV and Video
VD or USB	Cortoso, Ltd	Cortoso	Regular	\$208.52	\$409	256		11	40 pounds	40 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$197.28	\$429	257		11	11 pounds	11 pounds	TV and Video
with 5.1-channel DVD players, computer systems, and more	Cortoso, Ltd	Cortoso	Regular	\$224.87	\$489	258		11	48 pounds	48 pounds	TV and Video
il in/out	Cortoso, Ltd	Cortoso	Regular	\$167.73	\$329	259		11	66 pounds	66 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$155.89	\$339	260		11	84 pounds	84 pounds	TV and Video
ubwoofer	Cortoso, Ltd	Cortoso	Regular	\$183.49	\$399	261		11	33.1 pounds	33.1 pounds	TV and Video
ubwoofer	Cortoso, Ltd	Cortoso	Regular	\$243.27	\$529	263		11	34 pounds	34 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$244.72	\$480	264		11	44.1 pounds	44.1 pounds	TV and Video
id 1 subwoofer	Cortoso, Ltd	Cortoso	Regular	\$183.54	\$360	265		11	11.4 pounds	11.4 pounds	TV and Video
subwoofer and 1 A/V receiver	Cortoso, Ltd	Cortoso	Regular	\$167.73	\$329	267		11	19 pounds	19 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$294.54	\$889	268		11	41 pounds	41 pounds	TV and Video
VD or USB	Cortoso, Ltd	Cortoso	Regular	\$208.52	\$409	269		11	7.5 pounds	7.5 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$197.28	\$429	270		11	11.4 pounds	11.4 pounds	TV and Video
with 5.1-channel DVD players, computer systems, and more	Cortoso, Ltd	Cortoso	Regular	\$224.87	\$489	271		11	65 pounds	65 pounds	TV and Video
il in/out	Cortoso, Ltd	Cortoso	Regular	\$167.73	\$329	272		11	34 pounds	34 pounds	TV and Video
	Cortoso, Ltd	Cortoso	Regular	\$155.89	\$339	273		11	30.5 pounds	30.5 pounds	TV and Video

STEP4: Implementing Conditional Logic in an Expression using **STORE** table

FileHomeHelpTable tools

NameStore

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

Calendars

Structure

StoreKey

GeographyKey

StoreType

StoreName

Status

EmployeeCount

SellingAreaSize

OpenDate

CloseDate

Years Open

Lifetime Sales

Table: Store (306 rows)

1	693	Store	Contoso Seattle No.1 Store	On	17	462	4/12/2004 12:00:00 AM		22	\$22,994,925.1695
2	693	Store	Contoso Seattle No.2 Store	On	25	700	2/14/2004 12:00:00 AM		22	\$22,668,671.159
3	656	Store	Contoso Kennewick Store	On	26	680	2/12/2004 12:00:00 AM		22	\$22,470,746.1235
4	424	Store	Contoso Bellevue Store	On	19	455	3/1/2004 12:00:00 AM		22	\$21,237,087.116
5	677	Store	Contoso Redmond Store	On	33	560	4/2/2004 12:00:00 AM		22	\$22,642,835.091
6	575	Store	Contoso Yakima Store	On	47	1125	5/15/2005 12:00:00 AM		21	\$22,903,891.261
7	838	Store	Contoso Granger Store	On	22	500	6/16/2004 12:00:00 AM		21	\$22,127,419.7615
8	935	Store	Contoso Sunnyside Store	On	17	460	7/2/2004 12:00:00 AM		21	\$22,273,861.1955
9	941	Store	Contoso Toppenish Store	On	25	700	8/18/2004 12:00:00 AM		21	\$23,001,652.789
10	947	Store	Contoso Wapato Store	On	25	700	9/19/2004 12:00:00 AM		21	\$23,038,106.4665
11	813	Store	Contoso Cle Elum Store	On	26	680	10/20/2004 12:00:00 AM		21	\$23,092,694.339
12	930	Store	Contoso Snoqualmie Store	On	33	560	12/22/2004 12:00:00 AM		21	\$22,877,189.1725
14	825	Store	Contoso Fall City Store	On	47	1125	1/23/2006 12:00:00 AM		20	\$22,314,891.133
15	678	Store	Contoso Renton Store	On	22	500	2/24/2004 12:00:00 AM		22	\$22,856,635.2115
16	627	Store	Contoso Everett Store	On	17	462	1/25/2004 12:00:00 AM		22	\$22,745,221.8975
17	551	Store	Contoso Spokane Store	On	25	700	3/26/2004 12:00:00 AM		22	\$22,598,548.465
18	944	Store	Contoso Veradale Store	On	26	680	7/27/2004 12:00:00 AM		21	\$22,507,907.9575
20	824	Store	Contoso Englewood Store	On	33	560	9/29/2004 12:00:00 AM		21	\$22,514,748.192
21	950	Store	Contoso Wheat Ridge Store	On	47	1125	11/2/2004 12:00:00 AM		21	\$22,816,231.823
22	571	Store	Contoso Westminster Store	On	22	500	12/3/2004 12:00:00 AM		21	\$22,238,424.413
23	836	Store	Contoso Grand Junction Store	On	17	460	1/24/2004 12:00:00 AM		22	\$22,329,282.372
24	887	Store	Contoso New Castle Store	On	25	700	2/15/2004 12:00:00 AM		22	\$22,382,953.2435
25	839	Store	Contoso Greeley No.1 Store	On	25	700	3/16/2004 12:00:00 AM		22	\$22,463,731.9295
26	867	Store	Contoso Loveland Store	On	26	680	4/17/2004 12:00:00 AM		22	\$23,277,716.327
27	859	Store	Contoso Lafayette Store	On	19	455	5/18/2004 12:00:00 AM		22	\$21,925,370.068
28	809	Store	Contoso Castle Rock Store	On	33	560	6/19/2004 12:00:00 AM		21	\$22,851,184.749
30	829	Store	Contoso Fort Collins Store	On	22	500	8/21/2004 12:00:00 AM		21	\$22,397,040.109
31	839	Store	Contoso Greeley No.2 Store	On	17	462	9/22/2004 12:00:00 AM		21	\$22,975,399.041
32	877	Store	Contoso Milliken Store	On	25	700	10/23/2004 12:00:00 AM		21	\$22,972,604.726
33	799	Store	Contoso Berthoud Store	On	26	680	11/24/2004 12:00:00 AM		21	\$23,031,618.491
34	818	Store	Contoso Denver No.1 Store	On	26	680	12/2/2004 12:00:00 AM		21	\$22,861,756.6975
35	818	Store	Contoso Denver No.2 Store	On	19	455	10/3/2004 12:00:00 AM		21	\$22,688,007.184
36	818	Store	Contoso Denver No.3 Store	On	33	560	6/27/2003 12:00:00 AM		22	\$23,111,475.5885
37	803	Store	Contoso Boulder Store	On	47	1125	3/28/2003 12:00:00 AM		23	\$23,113,009.595
38	648	Store	Contoso Thornton Store	On	33	560	4/28/2003 12:00:00 AM		33	\$23,354,400.6335

Search

channelKey

DateKey

DiscountAmount

DiscountQuantity

Margin

Margin Percent

ProductKey

PromotionKey

ReturnsAmount

ReturnQuantity

SalesAmount

SalesKey

SalesQuantity

StoreKey

TotalCost

UnitCost

UnitPrice

Store

CloseDate

EmployeeCount

GeographyKey

Lifetime Sales

OpenDate

SellingAreaSize

Status

StoreKey

StoreName

StoreType

Years Open

64°F

Mostly sunny

Search

7:26 PM

5/21/2026